

CO-1 CASE STUDY

STUDENT CASE STUDY EXECUTION REPORT

Department: FED

Subject: DSA-2

Academic Year: I-III Semester (2025-2026)

Faculty Name: Ms. G.Mamatha

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Branch: CSE

Course Outcome: CO-1 AVL Tree

Case Study Topic:

Efficient Student Record Management using AVL Tree

Work Done Summary:

In this case study, student records are managed using an AVL Tree data structure. The AVL Tree maintains balance automatically during insertion and deletion operations, ensuring efficient searching and retrieval of student information. Student records are organized using Student ID as the key, allowing quick access to academic details while maintaining optimal performance.

Implementation Details:

1. Student records are stored using AVL Tree with Student ID as the key.
2. Each node contains Student ID, Name, Attendance, and Marks.
3. New student records are inserted into the AVL Tree.
4. Balance factors are calculated after each insertion.
5. LL, RR, LR, and RL rotations are performed whenever imbalance occurs.
6. Search operations retrieve student records efficiently.
7. Deletion operations remove records when necessary.
8. Inorder traversal displays student records in sorted order.
9. The balanced structure ensures efficient performance even for large datasets.

Result : Screenshots / Output

Final AVL Tree Structure:

```
└─ 103 (Charitha)
   └─ 105 (Rahul)
      └─ 110 (Priya)
      └─ 101 (Anjali)
         └─ 102 (Kiran)
            └─ 100 (Sara)
```

Inorder Traversal (Sorted Order):

```
100 (Sara)
101 (Anjali)
102 (Kiran)
103 (Charitha)
105 (Rahul)
110 (Priya)
```

Student Record Search:

```
Student ID: 103
Student Name: Charitha
```

AVL Tree Balanced Successfully

```
Search Time : O(log n)
Insertion Time : O(log n)
Deletion Time : O(log n)
```

1. Insertion Operation: **$O(\log n)$** due to AVL balancing.
2. Search Operation: **$O(\log n)$** because of height-balanced structure.
3. Deletion Operation: **$O(\log n)$** including rotation operations.
4. Inorder Traversal: **$O(n)$** as all nodes are visited once.
5. AVL Tree ensures efficient storage and retrieval of student records.

GitHub Link: <https://github.com/devicharithab-del/DSA-2-CO1>

Student Signature

Faculty Signature